How we measure: HALO and sondes

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- HALO research aircraft uses (drop)sondes
  - **measure** temperature, humidity, pressure, and horizontal wind
  - **calculate** integrated water vapor, …

- radiosondes

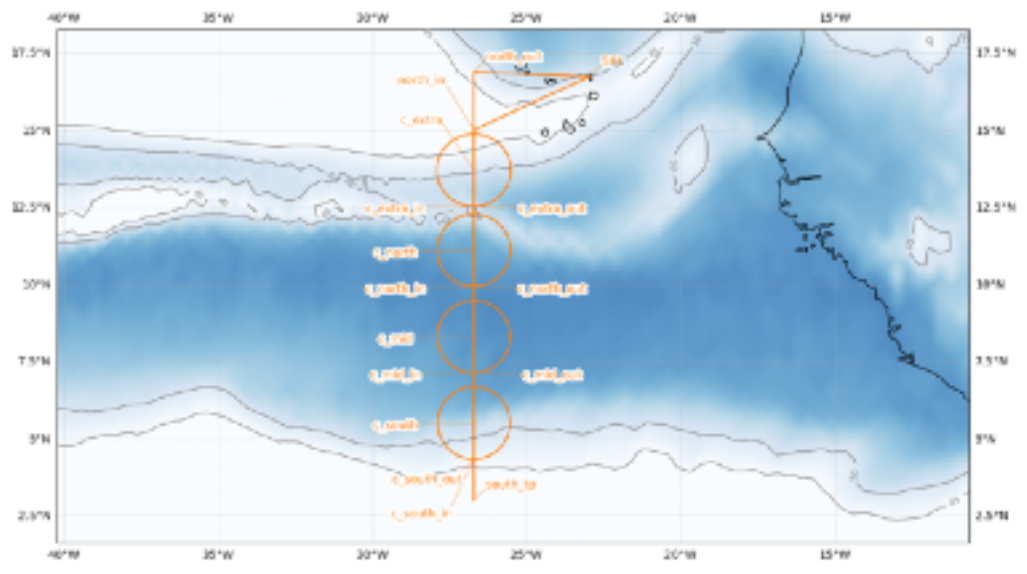
Belly Pod Section

a. Radiometer Suite
b. Water Vapour DIAL (WALES)
c. Cloud and Precipitation Radar
d. Thermal Imager
e. SWIRT

Tail Section

f. SWIRT
g. [illegible]
h. Dropsondes

Stevens et al. (2019)


40°W
35°W
30°W
25°W
20°W
15°W
17.5°N
15°N
12.5°N
10°N
7.5°N
5°N
2.5°N
c_north_out
c_north_in
c_north
c_mid_in
c_mid_out
c_south
c_south_out
south_to
c_south_in

HALO flight plan from 2024-08-21
(fun fact: Cathy was PI for this flight!)

why circles?

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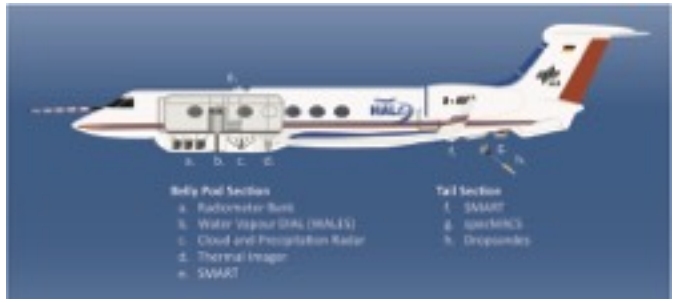
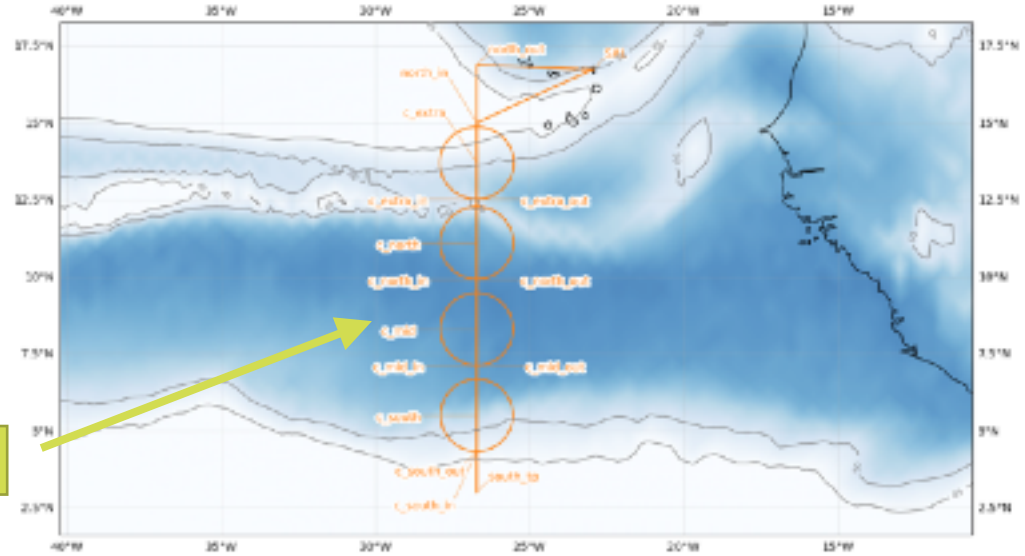


Stevens et al. (2019)

why circles?



HALO flight plan from 2024-08-21
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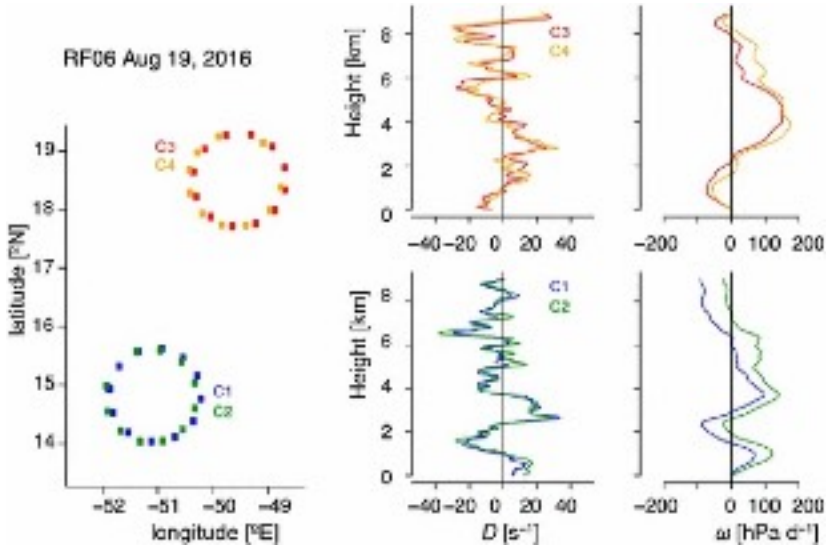
How we measure: campaigns

- Dropping sondes in a **circular** flight pattern:
  - large-scale vertical velocity

$$\omega = -\int D \, \mathrm{d}p$$

  - divergence of the horizontal wind

$$D = \nabla \cdot V_H$$



Bony, Stevens, et al. (2017)